

Vocabulary Master

Vocabulary Master is a full-featured offline web application for learning and mastering English vocabulary. It allows you to upload custom dictionaries (JSON or JSONL), study with interactive flashcards, take adaptive tests, track your progress, and manage personal word lists – all within your browser using IndexedDB.

Features

- **Flashcard System** – Flip cards to see definitions, examples, synonyms, and antonyms.
- **Dictionary Browser** – Browse thousands of words with alphabetical navigation, search, sorting, and pagination.
- **Adaptive Tests** – Create tests based on difficulty, genre, personal words, or trouble words. Question types: meaning, synonyms, antonyms, example-based, or mixed.
- **Progress Tracking** – Visual charts (score progress, mastery distribution), test history, and word-by-word mastery levels (New → Learning → Known → Mastered → Trouble).
- **Personal Word List** – Save words you want to focus on. Import/export your list, practice only your saved words.
- **Accessibility** – Font size controls, high-contrast mode, screen reader support, text-to-speech.
- **Dark / Light Theme** – Seamless theme switching with persistent preference.
- **Offline First** – All data is stored locally with IndexedDB. Works without an internet connection after the first dictionary load.
- **Performance Optimized** – Pagination, virtualised word grids, batch processing for large dictionaries (supports files >10 MB).

Getting Started

Prerequisites

- A modern web browser (Chrome, Firefox, Edge, Safari) with JavaScript enabled.
 - (Optional) Your own dictionary file in **JSON** or **JSONL** format. **Installation**
1. Download or clone the repository:

```
git clone https://github.com/UkeshMainali/Vocabulary-master.git cd vocabulary-master
```
 2. Open the index.html file in your browser.

For the best experience, serve the files with a local web server (e.g., `npx serve .` or `python -m http.server`).

First Launch

- On first load, you will be asked to upload a dictionary (JSON or JSONL) or use the sample words.
- The application processes the file and stores all words in IndexedDB. Large files may take a few seconds.
- Once imported, you can start using the flashcards, dictionary, tests, and progress pages.

Supported Dictionary Formats 1. JSON (single object)

```
json
{
  "example": {
    "word": "example",
    "ipa": "/ɪg'zæm.pəl/",
    "definitions": ["A representative instance or illustration."],
    "examples": ["This is an example sentence."],
    "synonyms": ["instance", "sample"],
    "antonyms": [],
    "genre": ["general"],
    "difficulty": 0.5
  }
}
```

Keys can also be the word itself: json

```
{
  "example": "A representative instance..."
}
```

2. JSONL (one JSON object per line)

Each line should contain an object with at least a word field. Example:

```
json
{"word": "dictionary", "ipa": "/ˈdɪk.ʃə.nə.ɹi/", "definitions": ["A reference book...", "..."], "examples": ["..."], "synonyms": [], "antonyms": [], "genre": [], "difficulty": 0.5}
{"word": "perseverance", "ipa": "/ˌpɜː.sɪˈvɪə.rəns/", "definitions": ["Persistence in doing something..."], ...}
```

Automatic Genre & Difficulty Assignment

If your file does not provide genre or difficulty, the application will:

- Analyse the word, definition, and context to assign one or more genres (e.g., academic, health, technology).
- Compute a difficulty score based on word length and definition complexity (easy, medium, hard).

Usage Guide

Flashcards

- Use **Previous / Next** to navigate.
- **Flip Card** or click the card to see the full details (definition, examples, synonyms, antonyms).
- **Shuffle** randomises the order.
- **Loop** toggles continuous cycling.
- **Add to My Words** saves the word to your personal list.
- **Speak** reads the word aloud.
- Use the filters on the left (difficulty, genre, “My Words Only”) to narrow down which words appear in the flashcard deck.

Dictionary

- **Alphabet** navigation (All, A–Z) filters words by starting letter.
- **Search** bar – type any part of a word.
- **Sort by** – alphabetical, difficulty, genre, or your mastery level.
- Each word card shows a preview of the definition and examples. Click **View All** or the card itself to open a detailed modal with all definitions, examples, synonyms, and antonyms.
- Pagination controls let you adjust the number of words per page (20, 50, 100, 200).

Tests

1. Configure your test:
 - **Test type** – Meaning, Synonyms, Antonyms, Example-based, or Mixed.
 - **Difficulty / Genre** filters.
 - **My Words Only** – test only words you have saved.
 - **Focus on Trouble Words** – words you previously answered incorrectly. ○ **Number of questions** (slider from 5 to 20, but limited by available words).
2. Click **Start Test**.
3. Answer each multiple-choice question. After each answer, you’ll see immediate feedback (correct/incorrect).
4. At the end, view your score, time taken, and detailed results.

5. You can **Review Answers** (see all questions with correct/incorrect indicators plus full word data) or **Retest Incorrect** to focus on the words you missed.

Progress

- **Overview stats** – total tests taken, average score, mastered words, current streak.
- **Score Progress chart** – line graph of your last 10 test scores.
- **Word Mastery chart** – donut chart showing distribution of mastery levels.
- **Test History** – list of recent tests with scores and times.
- **Word Mastery Levels** – browse all words filtered by mastery (All, New, Learning, Known, Mastered, Trouble). Click any word to see its full details.

My Words (Personal List)

- View all words you have added.
- Filter by mastery level.
- **Practice My Words** starts flashcard session with only your saved words.
- **Shuffle** randomises the order.
- **Export Words** – download your personal word list (with mastery data) as a JSON backup.
- **Import Words** – restore a previously exported list.
- Three practice modes: Flashcards, Quick Test, and a (future) Matching Game.

Accessibility Panel (bottom-right corner)

- **Speak** – reads the current flashcard word.
- **Larger / Smaller Text** – increases or decreases the base font size.
- **High Contrast** – overrides all colours to a high-contrast scheme (ideal for visual impairments). This setting persists across sessions.

Data Management

- **Re-upload Dictionary** – clears the current dictionary and lets you upload a new one (your personal words and progress are preserved unless you clear all data).
- **Clear All Data** – deletes everything (dictionary, personal words, test history, mastery, streak). Use with caution.

Technical Details

- **Frontend:** HTML5, CSS3, Vanilla JavaScript (ES6+).

- **Storage:** IndexedDB (via idb wrapper in script.js). Object stores:
 - words – the full dictionary.
 - personalWords – saved words with timestamp.
 - testHistory – test results and answers.
 - wordMastery – per-word mastery level and statistics.
 - streak – current learning streak.
 - appState – miscellaneous user preferences.
- **Charts:** [Chart.js](#) for progress and mastery visualisations.
- **Icons:** Font Awesome 6 (free CDN).
- **Performance:**
 - Batch processing of large JSONL files (500 words per batch).
 - Pagination for dictionary, personal words, and progress lists.
 - Event delegation for dynamic content.
 - Debounced search input.
- **Accessibility:** Semantic HTML, ARIA labels, keyboard navigation, high-contrast mode, dynamic font scaling, speechSynthesis API.

Project Structure

text

vocabulary-master/

```
├── index.html    # Main HTML structure (all pages, modals, upload UI)
├── styles.css    # Core styles (themes, layout, components)
├── extras.css    # Additional styles (alphabet nav, flashcards, modals, progress)
├── script.js     # Application logic (VocabularyApp class)
└── README.md    # This file
```

Configuration & Customisation

- **Default sample words** – built into `getComprehensiveSampleWords()` (~15 words). You can replace them with your own.
- **Page size** – default 50 words per page in dictionary. Change via the dropdown in the dictionary footer.

- **Theme** – automatically follows data-theme attribute. Override by editing :root and [datatheme="dark"] in styles.css.
- **High-contrast** – defined in extras.css under body.high-contrast. Modify colours as needed.

Troubleshooting

Issue	Likely solution
Dictionary upload fails or shows “No valid words found”	Ensure your file is either a valid JSON object (with words as keys) or a JSONL file with one valid JSON object per line, each containing a word field.
App gets stuck on “Loading...”	Open browser DevTools (F12) → Console. Look for errors. The most common cause is a corrupted IndexedDB. Use the “Clear All Data” button or manually clear site data in your browser settings.
Flashcards / tests show no words	Check your filters (difficulty, genre, “My Words Only”). Reset them to “All” and try again.
High-contrast mode does not persist after refresh	The setting is saved in localStorage. If you clear browser storage, it will reset.
Large files cause the browser to freeze	The application processes in batches, but extremely large files (e.g., >100 MB) may still impact performance. Consider splitting your dictionary or using a smaller subset.

Contributing

Contributions are welcome! Please follow these steps:

1. Fork the repository.
2. Create a feature branch (git checkout -b feature/amazing-feature).
3. Commit your changes (git commit -m 'Add some amazing feature').
4. Push to the branch (git push origin feature/amazing-feature).
5. Open a Pull Request.

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Acknowledgements

- [Chart.js](#) for beautiful charts.

- [Font Awesome](#) for icons.
- [Kaikki.org](#) for the inspiration of the JSONL dictionary format.

Happy learning!

If you encounter any issues or have suggestions, please open an issue on GitHub.